Practical 17 on Generalised Linear Models (GLMs)

Parkrun (https://www.parkrun.org.uk/) is a non-competitive event held across the UK (and now globally) every saturday where people are invited to run an approximate 5km route. There are now several hundred locations where these events are held.

The organisers of each event would like to have a good idea ahead of the event of the number of runners to assist with logistics. The organisers are all volunteers, and on a busy day require more people to help with 'barcode collection' which is used to calculate people's runtimes.

Data are avaialable from a sample of parkrun events held between 2014-2016. The data include the number of people who attended, grouped by age and sex. Additionally, data are available on meterological (weather) variables, as it has been hypothesised that the weather influences the number of runners in attendance. The varaibles of interest are as follows;

- **total** represent the total number of runners that attend each event
- **sm20-24** to **vw70_74** represent the number of runners in each category 'senior men (sm)', 'senior women (sw)', 'vetran men (vm)', 'vetran women (vm)', and the more specifc age category (in 5 years age bands)
- **loc** location of the event (4 locations)
- **date** date of the event
- **rh** relative humidity, which is a reliable indicator of whether it is raining. Higher values of rh indicate that it is raining
- **rhc** is relative humidity, centered on 70 (more than this meas rain is likely)
- **temp** temperature (degrees celcius)
- **tempc** is temperature, centered on 10oC
- **wind speed** and **wind direction** are included as variables, but are largely missing, so should be excluded from the analysis

Part 1: Data exploration

Task: Use appropriate graphical tools and summaries to explore the variables in the data and produce a table of baseline characteristics of parkrun attendance and other variables.

Hint: you should also consider the timing of the parkrun events. For example, what impact does the month of the year have on attendance?

Try not to spend too long on this in the session, we only have 90mins!

In [3]:
```
# load relevant libraries
library(ggplot2)
install.packages("sandwich")
library(sandwich)

setwd("~/Dropbox/Dropbox/Teaching-LSHTM/MSc - DataScience/SHDS-module/
#dir()
# load in the data - returns the dataset 'dat'
load("parkrun_attend_clean_2015.Rdat")
dim(dat)
```

```
The downloaded binary packages are in
        /var/folders/jp/f50wn8ds35b_c2ns4tld6v6h0000gr/T//RtmpEvdKd
j/downloaded_packages

539  39
```

In [19]:
```
head(dat)
table(dat$loc)
```

A data.frame: 6 × 39

	loc	date	rh	wet_bulb	temp	wind_speed	wind_direction	sm20_24	sr
	<chr>	<fct>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	
41	brockwell	01/04/2017	72.4	0	12.9	NA	NA	2	
43	crystalpalace	01/04/2017	81.3	0	11.0	12	210	1	
44	finsbury	01/04/2017	72.4	0	12.9	NA	NA	0	
48	kingston	01/04/2017	76.5	0	12.4	NA	NA	0	
67	brockwell	01/07/2017	72.2	0	16.1	NA	NA	2	
69	crystalpalace	01/07/2017	77.1	0	16.0	9	300	1	

```
    brockwell crystalpalace      finsbury      kingston
          130           136           128           145
```

In [20]:
```
print("These data should have not missing data")
summary(dat$tempc)  #
summary(dat$rhc)  #
```

```
[1] "These data should have not missing data"

   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
-12.000  -2.350   2.000   1.673   6.000  13.700

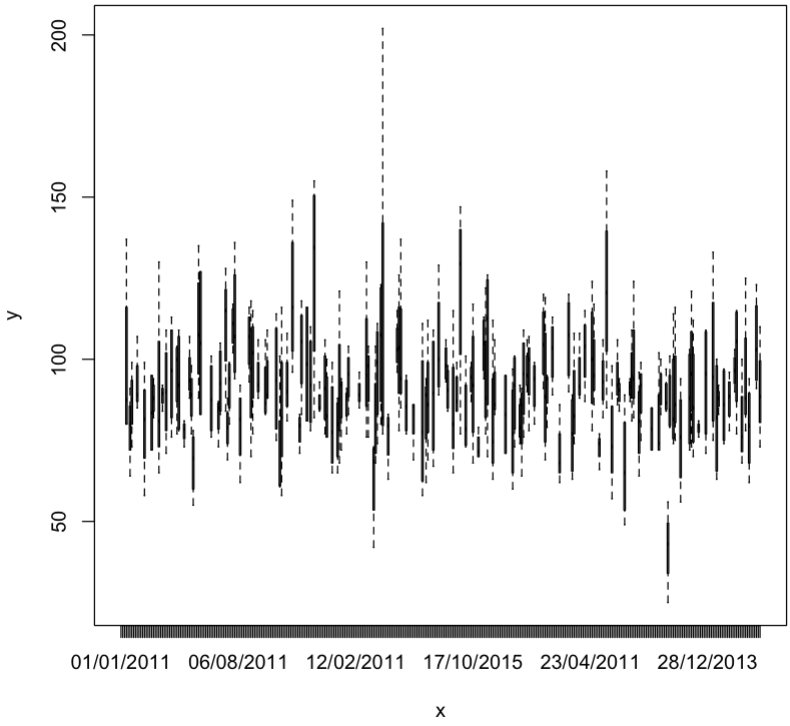
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
-21.100  -0.850   8.300   8.197  17.800  30.000
```

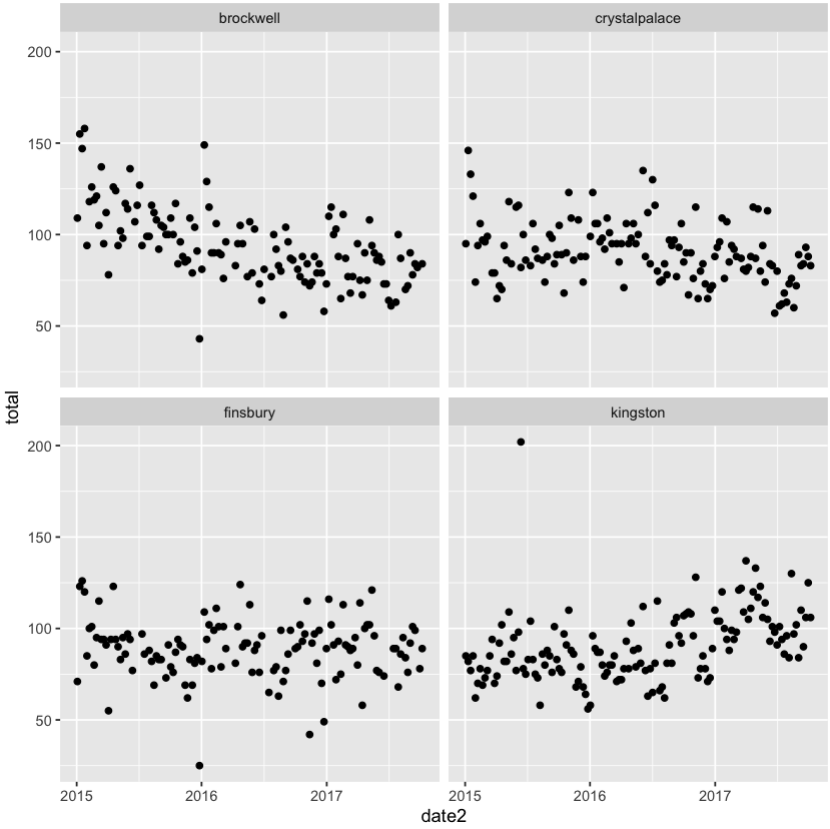
In [22]:
```
# plot the data - total number that attend

par(mfrow=c(1,1))
plot(dat$date,dat$total) # can't see much going on here...

# but, location might be important?
ggplot(dat,aes(x=date2,y=total,group=loc)) + geom_point() + facet_wrap

# decide on suitable summary stats...
```





Part 2: Univariable analysis.

Task: Use a Poisson GLM to estimate the effect of the weather and other variables on parkrun attendance.

In [1]:
```
# plot first...
# what is the relationship between temp and rh on total attendance. Is

# code here
```

Notes:

- A linear trend for temp & rh is not very strong if it is even present. Scatter plots indicate only a slight trend.
- Effect of rh on total is more obvious. It might be worth considering interactions between location and rh.

In [4]:
```
# part 2: GLMs - univariable

m1 <- glm(total ~ 1,family="poisson", data=dat)
summary(m1)

# this would be the 'default' model, and the effect of rh examined in
# Residual deviance:
# The residual deviance is the difference between the deviance of the
# the maximum deviance of the ideal model where the predicted values a
# the observed. Therefore, if the residual difference is small enough,
# fit test will not be significant, indicating that the model fits the

#  code here for further models
```

```
Call:
glm(formula = total ~ 1, family = "poisson", data = dat)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept) 4.510860   0.004515     999   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 1925.6  on 538  degrees of freedom
Residual deviance: 1925.6  on 538  degrees of freedom
AIC: 5339.9

Number of Fisher Scoring iterations: 4
```

Notes:

Univariable analysis:

Model	deviance	df.	AIC	p-value of Wald statistic
Null.	.	.	.	n/a
Year.	.	.	.	.
Loc.	.	.	.	.
Temp.	.	.	.	.
Rh.	.	.	.	.

In the regression model it is a 'Wald statistic' that is being applied to each explanatory variable to test for its statistical significance. (Note that any more information about the Wald statistic is not necessary for interpretation)

Notes...

Part 3: Checking assumptions of your model.

Task: Examine the models to see whether the data are Poisson distributed, or if an alternative model structure should be explored. Comment of the tests you used to make your decision.

Hint: some useful code are below for using robust standard errors. Note however, that either this option should be selected, **or** a quasi-poisson model specified, but not both!

In [5]:
```
pchisq(??, ??, lower.tail=FALSE)
```

```
Error in parse(text = x, srcfile = src): <text>:1:10: unexpected ','
1: pchisq(??,
             ^
Traceback:
```

Part 4: Multivariable analysis.

Task: Construct a multivarable model of *location, month, temp, and humidity* that could be used to predict parkrun attendance.

In [6]:
```
m2a <- glm(total ~ rhc + tempc + year + loc + month,family="??", data=
summary(m2a)  # no impact of rh

# output robust std errors
cov <- vcovHC(m2a, type="HC0")
se <- sqrt(diag(cov))
robust.SE <- cbind(Estimate= coef(m2a), "Robust SE" = se,
                   "Pr(>|z|)" = 2 * pnorm(abs(coef(m2a)/se),lower.tail
                   LL = round(coef(m2a) - 1.96 * se,3),
                   UL = round(coef(m2a) + 1.96 * se,3))
print(robust.SE)

# check assumptions
pchisq(m2a$deviance, m2a$df.residual, lower.tail=FALSE)  # p<0.01 idea

# so assumptions are a bit violated
# use package 'sandwich' to adjust the standard errors to 'robust' sta
```

```
Error in get(family, mode = "function", envir = parent.frame()): obj
ect '??' of mode 'function' was not found
Traceback:

1. glm(total ~ rhc + tempc + year + loc + month, family = "??",
 .     data = dat)
2. get(family, mode = "function", envir = parent.frame())
```

Part 5: Model predictions.

Task: The parkrun organisers have requested that you predict the expected numbers of runners when the weather has the following conditions;

For each of the locations (Brockwell, Crystal Palace, Finsbury, Kingston):

- date: 01/05/2017
- temp: 10oC
- rh: 70

Comment on your findings. In the practical we will compare findings across each group, are some age-sex groups more strongly influenced by the weather?

Hint: Using the function predict() will return the expectation of the number of runners. For a prediction interval, we suggest sampling around this value to generate the distribution of runners.

In [7]:
```
# predict
# here we are using "new data" to create this prediction

# set up the model and predictions
```

In [ ]:
```

```